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(1) Program / Project Title
Program: Bachelor of Science in Information Technology Project Title: Enhancing Intellectual Property Applications: PUP's E-IPMO Services for Academic Works
(2) Program/Project Proponent
Program Leader: Ma. Criselle Q. Trinidad Project Leader: Ma. Criselle Q. Trinidad <i>*Please attach curriculum vitae</i>
(3) Program/Project Co-proponent and Research Assistant/s
Paula Aliyah R. Cama Mark Reinier E. Garcia Raebv Lielmo V. Inocentes
(4) Program/Project Proponent's Department/College/Office
Program: Bachelor of Science in Information Technology College: College of Computer and Information Sciences
(5) Background and Significance
The Intellectual Property Management Office (IPMO) plays an important role in helping researchers at the Polytechnic University of the Philippines (PUP) protect the value of their intellectual property (IP) and transform research into marketable innovations by collaborating with academic, industrial, and investment sectors. It also manages technologies developed at the university for licensing or partnerships and requires that all academic works be submitted for copyright registration, with the university as a co-owner. Despite its role, the current IPMO system is hindered by inefficient, manual methods of handling large volumes of IP-related documents from faculty and students. This results in delays and limits the university's potential to expand its IP output. To address this, there is a need for an organized, automated system. The experience of the Intellectual Property Office of the Philippines (IPOPHL) in which it successfully adopted an automated process in 2012 proves that digitization can improve IP application efficiency and increase filings, as seen in the 2.5% rise in 2023 (Lim, 2024). Inspired by the success and improvement made by IPOPHL, the proposed capstone titled “<i>Enhancing Intellectual Property Applications: PUP's E-IPMO Services for Academic Works</i>” proposes the project “e-IPMO” which seeks to enhance PUP's IP application processes through a centralized digital platform. The system will streamline document submission, track applications in real time, and minimize losses and errors, resulting in a smoother experience for both applicants and administrators. e-IPMO offers significant benefits for both applicants and administrators of Polytechnic University of the Philippines (PUP). For applicants, it eliminates the need for multiple in-person submissions and visits, reduces errors, and speeds up processing through a structured digital platform which allows easier tracking of their application. For administrators, it enhances record-keeping accuracy and management, lessen overloaded workloads, improve productivity, and minimizes the risk of human error which allows improved decision-making with improved data processing and structured organization. Lastly, e-IPMO aims to boost productivity, ensure reliable documentation, and support national efforts to advance intellectual property development.

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(6) Objectives and Research Questions

The e-IPMO project aims to create a centralized digital system to improve the management of Intellectual Property (IP) applications at the Polytechnic University of the Philippines' Intellectual Property Management Office (PUP IPMO). Currently, the office relies on manual processes such as physical submissions and verifications which cause inefficiencies, delays, document misplacement, and a lack of transparency in tracking applications. The proposed system seeks to address these issues by streamlining the submission process, enhancing user experience, and improving data and document handling.

The project will involve building and implementing a web-based platform that will serve as a central hub for all application-related documents, records, and procedures. This platform will be accessible to students, faculty, and administrators, ensuring better transparency and efficiency. Ultimately, e-IPMO will support the formalization of IP rights and contribute to the university's development.

Specific Objectives

To develop a website to support PUP IPMO's needs. A website that aligns with PUP's IPMO vision and mission to promote intellectual property awareness, legal protection, university-student partnership, and innovation within the university.

Specific Objectives:

- To create a collaborative academic environment
- To secure and promote intellectual property rights
- To foster innovation and research
- To create a data-driven Analysis and Impact Monitoring

(7) Materials and Methods

A. Study Design

- Definitions
- Prospective/retrospective
- # of centers, randomized?
- Blinded, placebo, etc.
- Period of enrollment or chart review covers? Informed consent?
- Study Drugs, Device, or Intervention
- Randomization/Sampling (If appropriate)
- Instruments: Standardized Surveys, etc. (provide a description of each)
- Endpoints: the main thing that you are looking at

B. Study Population

- Who? Age, Sex, Race, Diagnoses if appropriate.
- Inclusion/Exclusion criteria
- Experimental group vs. Control group

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C. Assessment of Resources

Indicate how investigator will ensure that the study:

- Has sufficient access to the study population
- Has sufficient time to conduct and complete the study
- Has adequate qualified staff members to conduct the study
- Facility is adequate to conduct the study
- Staff has been adequately trained on the protocol and their specific research related duties

D. Study Procedures

Include a description of the study procedures (as they relate to the subject). Be sure to include (as applicable):

Plans for Recruitment

- The number and estimated length of each study visit
- Procedures and/or interventions that will be performed for each visit (a chart may be helpful)
- If a drug study, include instructions for administering drugs, handling instructions, and storage and disposal instructions
- For biological samples, explain how the samples will be collected, as well as storage, testing, and disposal methods
- For behavioral studies, identify the instruments being used and who will be administering the instrument (including their qualifications)
- If a survey study, include who created the survey, whether survey has been standardized, how survey will be distributed and returned, and how confidentiality will be maintained
- If chart review, indicate whether or not the materials will be obtained prospectively, or if the materials will come from previously existing specimens, records, or data. Explain what information will be used to identify potential human subjects for inclusion in research.

E. Data Collection

- How and what data will be collected (i.e. demographics, comorbidities, standardized tools etc).
- Reliability and validity of the research instrument

F. Data Analysis

- Sample Size Considerations
 - Power analysis based on previous studies or exploratory study?
 - Justifying the sampling procedure
- Statistical Methodology
 - Comparisons to be made and statistical tests to be used for the comparisons.

(8) Safety and Monitoring Plan

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The research involves the collection of survey feedback from two groups: the administrators and end users. Using stratified purposive sampling to ensure representative input from both perspectives. The survey may be implemented online or through printed forms. The nature of the study poses minimal to no risk to participants. No sensitive personal or psychological data will be collected, and all questions will relate strictly to user experiences, preferences, and suggestions for website improvement. There are no anticipated physical, emotional, or psychological harms involved in the study. Although the study poses minimal to no risk to participants. The researchers will take precautions to ensure participants' safety and as well as data security. The precautions are the following: 1. Voluntary Participation and Informed Consent---participants will be given clear instructions regarding the survey and sufficient information about the study. Information and instructions include the purpose, their role, and their rights. Importantly, participants must provide informed consent before participating. 2. The right to withdraw---participants may withdraw at any point without any negative consequences. For digital survey forms, it is provided before the submission. For paper forms, participants may choose not to submit/answer the form. 3. Survey Neutrality---all survey questions will be formulated in neutral, non-leading language to avoid imposing discomfort and biased questions on the participants. For Data Safety, the CIA principles will be observed and serve as a guide in all data handling. Participants have the option to cast their profile as anonymous while answering the questionnaire. All the answers are stored securely (encrypted drives/locked cabinets). Only the authorized personnel, e.g., researchers, can access the information provided by the participants. The information will only be kept for the approved research period, then securely destroyed.
(9) Limitations

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This study focuses on creating and using the e-IPMO system at the Polytechnic University of the Philippines (PUP) – Main Campus and its branches. The system is designed to make the Intellectual Property Management Office (IPMO) more efficient by streamlining the process for students and faculty to apply for intellectual property (IP). However, it only handles tasks related to submitting, tracking, and managing IP applications. The system does not deal with issues like duplicate studies, disputes over authorship and ownership, or plagiarism. Additionally, the e-IPMO system's success depends on how actively both users and administrators engage with it; limited participation may lessen the benefits of centralizing and automating processes. Another limitation is that the system is only for internal use within PUP and is not meant for outside institutions or third-party connections. Since the system manages sensitive user information, strong security measures are crucial. However, because cybersecurity threats and institutional needs change over time, the system might need updates and improvements in the future to keep user data secure and maintain its effectiveness.
(10) Ethical Considerations

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A. Informed Consent (Applies to studies using human subjects)

The consent and legal rights of both the administrators and end users are respected throughout their navigation on the website. Any participation requiring consent and agreement will be clearly presented on the screen prior to proceeding with any activities. Consent is confirmed by selecting the checkbox before continuing with any questionnaires or surveys. Users have the right to refuse or decline any permission they believe infringes upon their privacy or security. Furthermore, participating in these activities is not only necessary but also contributes to the ongoing development and improvement of the university's website, helping researchers enhance user experience and functionality. Rejecting consent will result in the user being immediately exited from the website without any negative consequences. Only the researchers will be involved in gathering any form of consent regarding this study, either a digital or physical survey and questionnaire. The survey reflects a brief explanation of the study's purpose and how the responses will be used, ensuring data governance.

B. Risks and Side Effects (Applies to studies using human subjects)

This document outlines the potential risks and side effects associated with participant involvement in the study, alongside the measures in place to ensure safety, confidentiality, and the well-being of all participants. The potential risks and side effects of participating in this study have been carefully considered to ensure the safety and well-being of participants. The review includes risks about emotional distress, privacy and confidentiality risks, physical discomfort, and technical issues.

Emotional Distress

Participants may experience discomfort when responding to sensitive questions. To mitigate this risk, study questions are designed with neutral wording to prevent leading or suggestive responses. Participants will be informed prior to participation that they can withdraw at any moment without facing any negative consequences.

Privacy and Confidentiality

We acknowledge the inherent risks associated with privacy and confidentiality, particularly with digital data. Despite robust security measures, minimal risk of confidentiality breaches exists. All responses will be securely stored, with access strictly limited to authorized personnel. Participants will be assured that their identities will remain confidential throughout the study process.

Time Commitment

Participation may require a significant time investment, which could disrupt participants' daily routines. To address this, participants will receive clear information regarding the expected duration of commitment prior to enrollment. They will have the freedom to withdraw from study should the time commitment become burdensome.

Physical Discomfort

Participants may experience minor discomfort due to prolonged sitting or extended use of technology. They will be encouraged to take breaks as needed and have the option to discontinue participation if discomfort becomes excessive.

Technical Issue

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Participants may encounter technical malfunctions or server issues during their involvement. To address this, technical support will be readily available. Participants will be informed that they can exit the study if any technical problems adversely affect their experience.

To minimize the above risks, the study will implement secure data storage practices and stringent measures to protect privacy. Comprehensive information will be provided during the consent process, highlighting participants' rights, including the ability to withdraw at any time. Furthermore, a debriefing session will be conducted post-study to inform participants about the research purpose and the intended use of collected data.

The safety and well-being of our participants are of utmost importance. By identifying potential risks and implementing appropriate mitigation strategies, we aim to conduct this study ethically and responsibly. We seek ethical clearance to proceed with this important research.

C. Benefits to Subjects (Applies to studies using human subjects)

The participants in this study, including students, faculty members, and IPMO staff, will benefit from the development of a more accessible and efficient system for processing intellectual property applications. By participating, they contribute to the improvement of a platform that will help streamline submissions, reduce manual paperwork, and promote better awareness of intellectual property rights within the academic community. The project aims to create a more user-friendly experience that supports the protection and recognition of academic works at PUP.

D. Costs to Subject (Applies to studies using human subjects)

There are no significant costs or risks involved for participants in this study. Participation may require a small amount of time to answer surveys or take part in interviews, but this will be kept brief and voluntary. No financial contributions or sensitive procedures are involved, and all information gathered will be handled confidentially and used only for the purposes of improving the e-IPMO system.

E. Compensation to Subject (Applies to studies using human subjects)

Participants will not receive any monetary compensation for their involvement in the study. However, their contributions will play an important role in shaping a digital service that can benefit the entire university community. Acknowledgment of their input may be included in the final project documentation, with their consent, as a form of appreciation.

F. Provisions for vulnerable subjects (Applies to studies using human subjects)

If any vulnerable subjects (such as students who may feel pressured to participate) are included in the study, clear steps will be taken to ensure their voluntary consent. Participation will be optional, and they may withdraw at any time without consequence. All data will be well-protected to ensure their privacy, and the study will follow ethical guidelines to ensure their safety and comfort throughout the process.

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- G. Subject Privacy and Data Confidentiality (Applies to studies using human subjects)
- 1. Privacy of Participants
 - a) Describe appropriate provisions to protect the privacy of the subjects.
 - 2. Confidentiality of Data
 - a) Provide a clear description of how the data will be disseminated. Outline the sharing of data with others outside of the institution, and include provisions for maintaining confidentiality. Additionally, describe how the results of the data will be used (i.e. presentations at professional organizations, submission to professional journals).
 - i. Will data be identified?
 - ii. How will Data be kept Secure (where will data be stored)
 - iii. Who will have access to the data?
 - 3. Plan for Record Retention and Disposal
 - 4. Limits to Confidentiality

(11) Plan for Dissemination of Findings

The results of this study will be formally presented during the capstone defense at the Polytechnic University of the Philippines-Manila and shared with the IPMO office for potential implementation. A copy of the final report will be submitted to the university, and a system demonstration may be offered to the IPMO office for potential use or feedback. The capstone project may be submitted for inclusion in university research to raise awareness about intellectual property rights and promote the use of digital tools in academic work. If approved, the system could be recommended for adoption to enhance IP services in the university.

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(13) Appendices

This refers to research instruments, rating scales, consent forms, etc.

Signature	(14) Prepared by :	(15) Endorsed by:
Name of proponent	Ma. Criselle Q. Trinidad	Janelle Kyra Sagum
Designation/position	Researcher	Research Adviser
Date		